Kernix 0.5.0 / Performance brief

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

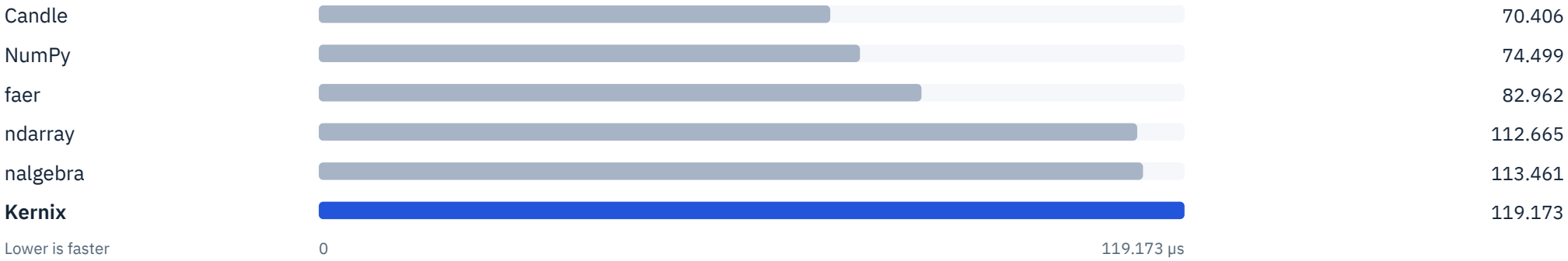
09 workloads

40 observations

Source precision retained

Matrix multiplication

128 × 128 · 128 × 128 · f64 · 1 worker · allocated output · self product



How to read this report

Allocation-inclusive API latency unless marked reused output. Peer medians are observations.

Charts round to 6 significant digits (≈). The exact-value appendix preserves all recorded digits.

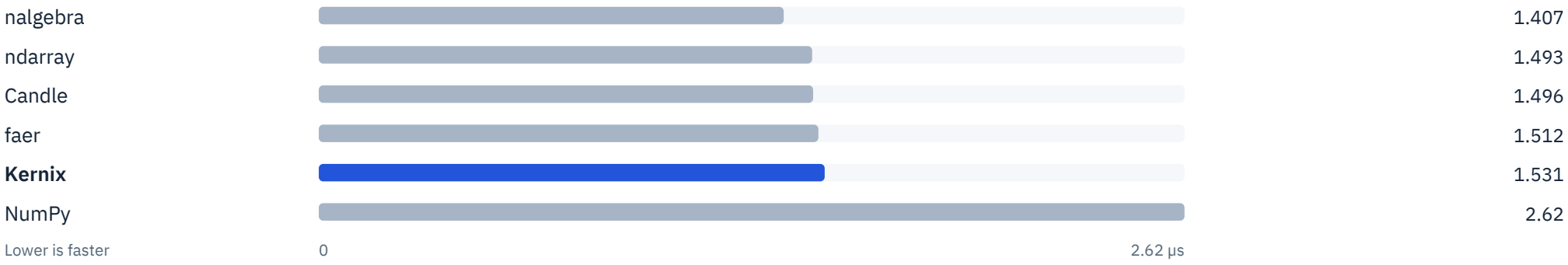
Kernix 0.5.0 / Matrix multiplication

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Matrix multiplication

Per-call median

32 × 32 · 32 × 32 · f64 · 1 worker · allocated output · self product

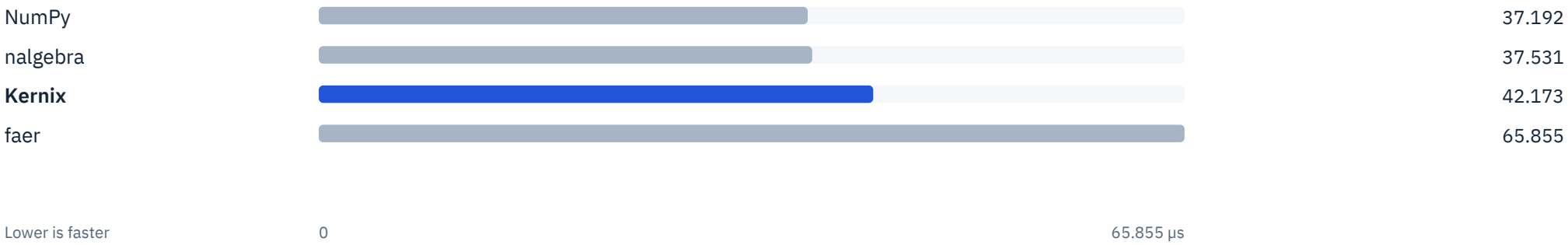


Kernix 0.5.0 / Decompositions & solvers

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Thin QR

96 × 32 · f64 · 1 worker · allocated output



Thin QR

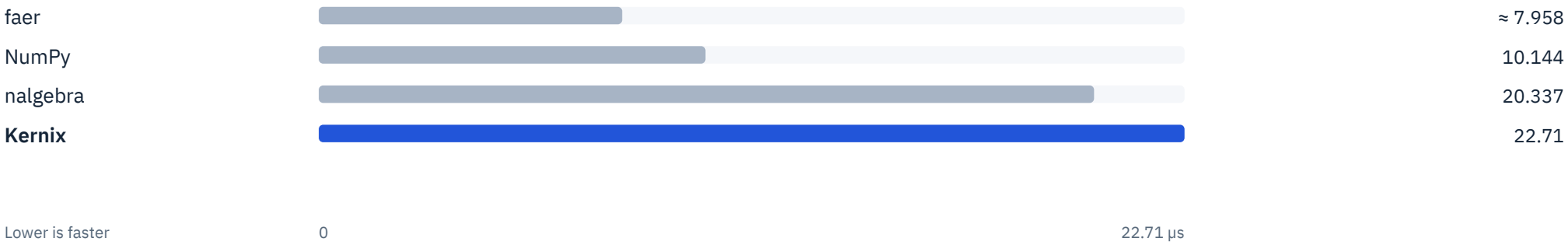
256 × 64 · f64 · 1 worker · allocated output

Kernix 0.5.0 / Decompositions & solvers

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Cholesky factorization

64 × 64 · f64 · 1 worker · allocated output



Kernix 0.5.0 / Training & autodiff

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Record + backward

Per-call median

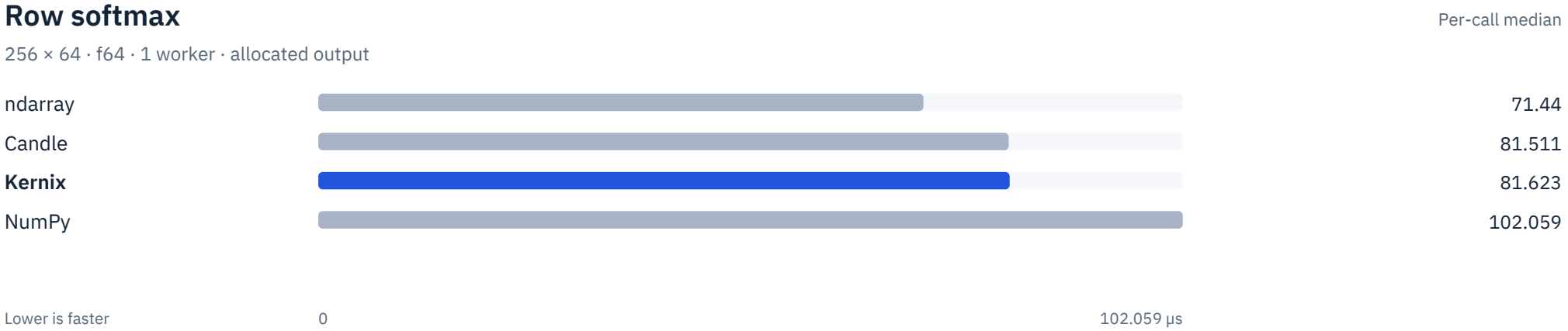
sum(A · W), 32 × 16 · 16 × 8 · f64 · 1 worker · allocated output · self product

Kernix 0.5.0 / Statistics & activations

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Row softmax

256 × 64 · f64 · 1 worker · allocated output



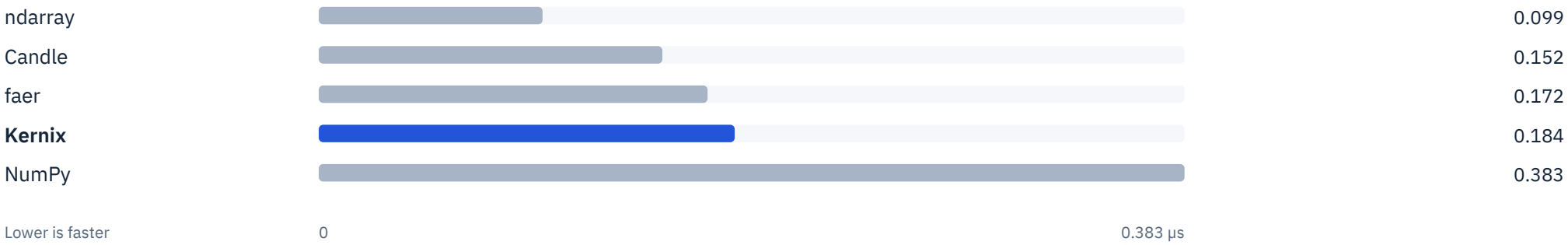
Kernix 0.5.0 / Tensor construction

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Create zeros

Per-call median

64 × 64 · f32 · 1 worker · allocated output



Create ones

Per-call median

64 × 64 · f32 · 1 worker · allocated output

Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / Kernix 32 × 32 · 32 × 32 · f64	1531	1.531
Matrix multiplication / ndarray 32 × 32 · 32 × 32 · f64	1493	1.493
Matrix multiplication / nalgebra 32 × 32 · 32 × 32 · f64	1407	1.407
Matrix multiplication / faer 32 × 32 · 32 × 32 · f64	1512	1.512
Matrix multiplication / Candle 32 × 32 · 32 × 32 · f64	1496	1.496
Matrix multiplication / NumPy 32 × 32 · 32 × 32 · f64	2620	2.62
Matrix multiplication / Kernix 128 × 128 · 128 × 128 · f64	119173	119.173
Matrix multiplication / ndarray 128 × 128 · 128 × 128 · f64	112665	112.665
Matrix multiplication / nalgebra 128 × 128 · 128 × 128 · f64	113461	113.461
Matrix multiplication / faer 128 × 128 · 128 × 128 · f64	82962	82.962

Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / Candle 128 × 128 · 128 × 128 · f64	70406	70.406
Matrix multiplication / NumPy 128 × 128 · 128 × 128 · f64	74499	74.499
Thin QR / Kernix 96 × 32 · f64	42173	42.173
Thin QR / nalgebra 96 × 32 · f64	37531	37.531
Thin QR / faer 96 × 32 · f64	65855	65.855
Thin QR / NumPy 96 × 32 · f64	37192	37.192
Thin QR / Kernix 256 × 64 · f64	502963.99999999994	502.96399999999994
Thin QR / nalgebra 256 × 64 · f64	380362	380.362
Thin QR / faer 256 × 64 · f64	426510	426.51
Thin QR / NumPy 256 × 64 · f64	346403	346.403

Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Cholesky factorization / Kernix 64 × 64 · f64	22710	22.71
Cholesky factorization / nalgebra 64 × 64 · f64	20337	20.337
Cholesky factorization / faer 64 × 64 · f64	7958.000000000001	7.958000000000001
Cholesky factorization / NumPy 64 × 64 · f64	10144	10.144
Record + backward / Kernix sum(A · W), 32 × 16 · 16 × 8 · f64	2954	2.954
Record + backward / Candle sum(A · W), 32 × 16 · 16 × 8 · f64	3982	3.982
Row softmax / Kernix 256 × 64 · f64	81623	81.623
Row softmax / ndarray 256 × 64 · f64	71440	71.44
Row softmax / Candle 256 × 64 · f64	81511	81.511
Row softmax / NumPy 256 × 64 · f64	102059	102.059

Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Create zeros / Kernix 64 × 64 · f32	184	0.184
Create zeros / ndarray 64 × 64 · f32	99	0.099
Create zeros / faer 64 × 64 · f32	172	0.172
Create zeros / Candle 64 × 64 · f32	152	0.152
Create zeros / NumPy 64 × 64 · f32	383	0.383
Create ones / Kernix 64 × 64 · f32	183	0.183
Create ones / ndarray 64 × 64 · f32	137	0.137
Create ones / faer 64 × 64 · f32	284	0.284
Create ones / Candle 64 × 64 · f32	216	0.216
Create ones / NumPy 64 × 64 · f32	1339	1.339